

Chapter 1

Results

This chapter presents the results from applying the counterfactual analysis methodology to a path-based knowledge graph recommender system. We start by examining the initial recommendations generated by the recommender system for a random user. Following this, we explore the outcomes of our counterfactual analysis to understand how different attributes affect product recommendations.

1.1 Case Study

To provide a concrete example of our counterfactual analysis methodology process, we will conduct a case study. This case study will illustrate the application of our methodology, demonstrating its effectiveness and potential insights.

1.1.1 Initial Recommendations

To initiate our analysis, we first examined the top 10 recommendations generated by the CAFE recommender system engine for a specific user (user 0). These recommendations, along with their associated paths and log probabilities, provide insight into the system's decision-making process. Below, we present these recommendations and discuss their implications.

Top 10 Recommendations for User 0:

1. ([0, 261, 849],
[-4.5635014, -3.4569123],
["mentions", "rev_described_by"])

2. `([0, 9826, 123],
[-5.4080114, -3.6974714],
["mentions", "rev_described_by"])`
3. `([0, 14974, 396],
[-5.5097566, -3.6836069],
["mentions", "rev_described_by"])`
4. `([0, 6797, 528],
[-5.648406, -3.7231576],
["mentions", "rev_described_by"])`
5. `([0, 856, 455, 330],
[-2.7392018, -1.455907, -3.5228138],
["purchase", "also_viewed", "rev_also_bought"])`
6. `([0, 856, 479, 626],
[-2.7392018, -2.2539396, -3.566442],
["purchase", "also_viewed", "rev_also_bought"])`
7. `([0, 856, 4350, 624],
[-2.7392018, -8.304826, -4.138239],
["purchase", "described_by", "rev_described_by"])`
8. `([0, 856, 13745, 241],
[-2.7392018, -8.540054, -4.213147],
["purchase", "described_by", "rev_described_by"])`
9. `([0, 856, 3327, 793],
[-2.7392018, -8.5651655, -4.2710376],
["purchase", "described_by", "rev_described_by"])`
10. `([0, 856, 3, 45],
[-2.7392018, -0.89929277, -3.0066423],
["purchase", "produced_by", "rev_produced_by"])`

Analysis of Recommendations:

Path Types: We observe four distinct path types in these recommendations:

- 'mentions' → 'rev_described_by' (4 instances)

- 'purchase' → 'also_viewed' → 'rev_also_bought' (2 instances)
- 'purchase' → 'described_by' → 'rev_described_by' (3 instances)
- 'purchase' → 'produced_by' → 'rev_produced_by' (1 instance)

The probability scores (represented as negative log-likelihoods) [placeholder for logo probs definition] vary significantly across the recommendations. Lower absolute values indicate higher probabilities.

In the following sections, we will explore how modifications to these paths and attributes might alter the recommendation outcomes through our counterfactual analysis.

1.1.2 Counterfactual Analysis for User 0's Recommendations

Below is an example of a recommended path that user 0 is recommended product 45 based because of the brand of their previous purchases:

```
([0, 856, 3, 45],
 [-2.7392018, -0.89929277, -3.0066423],
 ["purchase", "produced_by", "rev_produced_by"])
```

That with IDs converted to their corresponding real name for the attribute, it represents:

```
user_0 → purchases → product_856 → produced_by → brand:Maybelline
      → has_produced → product_45
```

After removing the outlier nodes and determining the communities, as illustrated in figure 1.1 on the following page, we select the first-layer attributes that are connected to a product that, in turn, is connected through the attributes of the recommended product. This selection is made for all entities connected to the recommended product. Based on these attributes, counterfactual scenarios are constructed as described in the previous section, and the recommender engine evaluates their plausability.

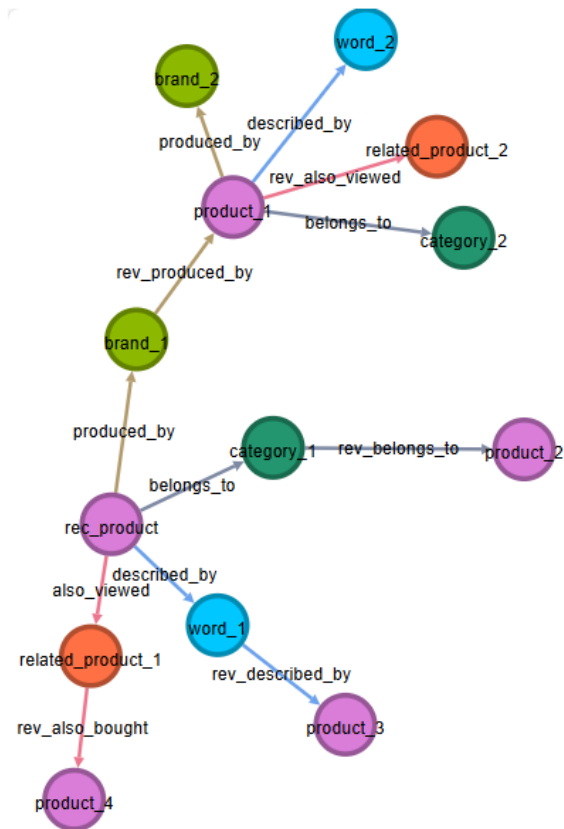


Figure 1.1: Demonstration of First Layer Attribute Selection

We consider the score of the path leading to the 10th recommended product as the minimum acceptable threshold. If a counterfactual path achieves this score, it indicates a scenario where the recommender engine assigns a score suggesting that the user is likely to appreciate the recommendation.

figure 1.2

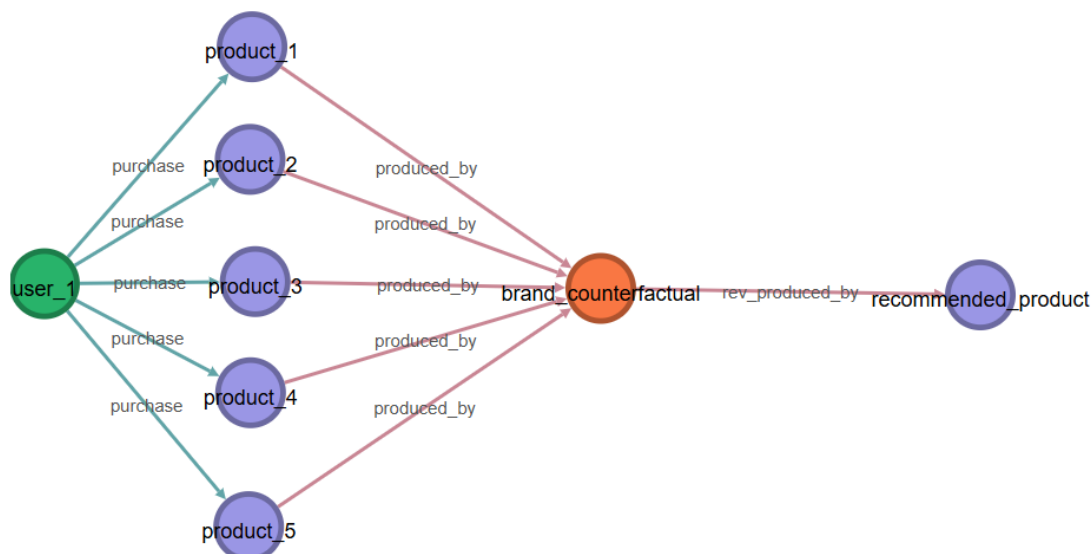


Figure 1.2: An Example of the Construction of Counterfactual Paths for an Attribute Evaluation

The original attributes of the product in this recommendation are:

Words the product is described by:

marker	lipstick	touch	fabulous
anybody	liquid	satisfy	full
iti	wink	shine	precisely
bee	coloursensational	mine	morning
gosh	usage	stiff	point
drawing	downwards	sun	mail
record	kinda	productmy	beyond
gloss	eh	lipcolor	yes
greatit	z	hubby	ordered
rim	berry	regular	remind
lightit	part	turn	immediately

colorstay	wonder	bare	shipping
pop	fricken	chemical	running
ahead	somewhat	dissapointe	fix
sparkle	tint	tacky	delivery
pleasedi	delicious	reapp	intended
rosy	colour	sigh	lasting
nude	outside	itll	bag
folks	ago	ugly	moist
edge	sorta	eos	fade
insult	onto	periodical	amaze
coloring	rather	cost	pieces
itit	medicate	fact	ready
experienced	fuscia	luv	hurt
ohh	matte	toilet	recieve
lipline	taste	mouth	productit
cherry	maybelline	application	properly
stung	assume	complaint	okay
senior	effect	waist	bright
cracked	loving	send	vibrant
indicative	nature	luck	raspberry
wine	palest	exactly	sting
stain	cooltone	else	uncomfortable
lid	watery	kiss	pen
arrive	slightly	took	eat
person	tempt	bother	koolaid
tube	hooked	placed	matter
information	mom	close	overit
name	pigment	must	broken
dayin	medium	alot	orange
buff	gave	m	sell
trash	total	wish	scrub
least	dobut	appear	means
lipstickthe	layer	stroke	certainly
weird	place	peel	obviously
waste	goodnot	drink	cake

putt	understate	practical	liner
picture	chapstick	inevitable	guess
eaten	low	making	sharpie
reorder	future	true	cap
tip	gorgeous	defiant	tend
olay	balm	sum	darkish
judge	damage	n	working
manner	mention	reason	open
lipstain	applied	saturday	thumb
doit	seller	seal	wild
star	inflexible	comfortable	napkin
convenient	clump	poor	paper
mys	bottleit	funky	crayola
inexpensive	roll	fall	barely
feltpip	pick	faith	fruit
unfortunately	yo	pinkre	
soon	magic	worst	

Brand of the product is:

Maybelline

Categories the product belongs to:

Makeup	Lips	Lip Stains	Beauty
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Related Products that have been bought together, also bought, or also viewed with this product:

B006UEUJII	B00HQLC91E	B00303STIY	B00I4DRRXI
B004O3YCQ2	B0045KLPI2	B002YG5IZA	B00BEWRDUI
B00HF6S88I	B0051TSKTS	B00CTONUL6	B00HWT32SE
B004Y9LAW0	B004EFHY0Q	B002YGB8D6	B002YGBNSG
B00HUAOE2S	B006UEUJV0	B00HF7I6ZM	B00H3DAWO6
B00EYZR3E8	B00HWSNF00	B006HCJXBM	B00A86SDLO
B00383O43U	B006UEUH XK	B0082D9R1K	B006UET4XE
B00IR6HW5A	B00F9AALTG	B00HWSNAPU	B002YGDCNU

B004ECH328	B004BCXAM8	B005CYK9HS	B00F0KWBCU
B004KOSZTK	B00CTOLDEC	B007GMJ8Z8	B006XLA6U4
B006UEUIJS	B00HUAPZK8	B006UEUHLC	B009HVMVLY
B006UEUIUC	B004EF6DQC	B0051TSL10	B00FZF3C46
B00BSIPWS8	B009FAPVJG	B004EF8IGU	B002YG6048
B006K5J3WA	B004Y9LNEK	B00H2MSAVK	B00JULSPA2
B006UEUK82	B004679FRM	B00HDE0Q00	B002YGF5FI
B00AM3Z2XU	B004O3YCCL	B005CYK6MG	B00383Q11I
B00383KCCW	B001TK1I4M	B004O3ZPMW	B002YGF5I0
B008GOR6O0	B004EFJW06	B00HWT1A8S	B005OGSO3K
B00FORNVGO	B0045KUO8E	B008B2Z94Q	B004ECGPMM
B008JAPA7Q	B002YGB9JY	B004Y9JVWG	B006XM9M58
B006UEUGWM	B004ECC24W	B004Y9LMN2	B005CYJKAU
B00F0KHUDD	B004CKWWOQ	B0082D9RYM	B004ECIW3C
B005CYKJHI	B00HUAO7ZM	B006UETDR6	B0030GTYM6
B0051TSLOW	B006UEUI8E	B004Y9LM8C	
B006UEUH80	B00HDE1672	B00HU8W2BK	
B0082D9SUK	B0057WJ1N8	B002YGF4QS	

And the plausible attributes approved by the recommender engine are as follows:

BRAND: If the product had one of these as **BRAND**, it would still be recommended through that brand:

HEY	loel	Physicians Formula	Neutrogena
leegoal	Kevyn Aucoin	Epower Mall	Sigma Beauty
e.l.f. Cosmetics	SHANY Cosmetics	Starry	Crazy Cart
Coastal Scents	Max Factor	COVERGIRL	Ailunce
Mehron	NYX	L'Oreal Paris	Maybelline
Jordana	Godefroy	Lime Crime	divaderme
Bonamart	Sleek	Sparkling Beauty	Cover Girl
MISSHA	Essie	Benefit Cosmetics	Wonder Wedges
Hard Candy	Kingmys	Revlon	

Related Products: If any of these **Related Products** were connected to the recommended product, the product would still be recommended through that related product:

B001H270RQ	B005BF1M10	B004BCX8B6	B0013L3XMM
B004Y9GTYE	B002ULRAQE	B001KYNZM0	B00FAEOCP0
B00B1ZRTP2	B004Y9M81W	B00303STIY	B001IB0NOS
B00EL1HZ34	B006UET4XE	B00112BUBO	B006UETDGC
B00BMW24TU	B000CECSAO	B00150LT40	B00KVBZAXK
B002UL5ZKC	B008FX7B9M	B004B3YC9M	B008GOR6O0
B00HF6SJVY	B000US0804	B005OZH88I	B004Y9GV60
B00B1DRA94	B001KYQA1S	B002UL10M4	B00FORNVGO
B008WG7X8Q	B006UEUI8E	B009ZB6MKM	B005UKZAJG
B003K16T06	B0030GTYM6	B006OZEYR0	B003FKSVMG
B002ZTT9II	B002ISV7B8	B004BCXAM8	B009WI3RZQ
B00HWSNF00	B001HKR6WM	B004Y9LM8C	B005EIIEQA
B002WTC38O	B004Y9LAW0	B0048KSGZO	B006OZF3DY
B009K7X26I	B004X1QY20	B00DHQCJTO	B004Y9GVZG
B009HVMVLY	B0013TM9UQ	B004M7WKKK	

WORDS: No word was verified, as if it was connected to the recommended product, the product would be recommended through that word.

The plausible scenarios work in favor of the explainability by demonstrating how changes in specific product attributes, such as the brand or ingredients, could still lead to a strong recommendation for the user. This enhances the system’s transparency by showing that the recommendation is not solely dependent on a single brand or fixed set of attributes but can adapt to user preferences, as long as the core characteristics of the product align with the user’s needs.

Counterfactual Analysis Results and Their Interpretation

For a given path to undergo counterfactual analysis, the counterfactual scenarios derived from validated neighbors are evaluated by the recommender system engine. The plausible attributes, according to the engine, are presented in four groups: brand, category, word, and related products. The revelations of each group of attributes are as follows:

- **Brand Influence:** If the recommended product had been associated with a different brand within the same category and still met the threshold score, this implies that brand substitutability exists within certain user preferences, indicating flexibility in brand choice.

- **Word Influence from Reviews:** Altering descriptive attributes (for example adding “animal cruelty free”, “organic”) and finding that the product still surpasses the score threshold suggests the potential descriptive qualities that the user will find favorable for the recommended product.
- **Connectivity and Exposure:** Introducing hypothetical connections to recommended product and products (e.g., often bought together or viewed together) and observing their impact on the recommendation score could highlight potential for cross-selling strategies.

1.1.3 Key Marketing Findings and Enhancing Marketing Strategies

This section outlines the marketing insights derived from our counterfactual analysis:

1. **Brand Associations:** The wide range of brands identified in the counterfactual analysis suggests that the recommended product has broad appeal across different market segments.
2. **Key Descriptors:** Words like "stays" and "eyelash" provide insights into the product's key selling points, which could be emphasized in marketing materials.
3. **Related Product Opportunities:** The identification of related products that influence recommendations presents cross-selling and bundling opportunities.

These insights can be utilized to enhance marketing strategies in several ways:

1. **Targeted Advertising:** By understanding the diverse attributes that lead to a recommendation, marketers can create more targeted and effective advertising campaigns.
2. **Product Positioning:** The variety of associated categories and brands allows for flexible product positioning to appeal to different customer segments.
3. **Content Marketing:** The identified key words and categories can inform content creation, ensuring that product descriptions and marketing materials align with the attributes that drive recommendations.
4. **Cross-Promotion:** The related products identified in the analysis can guide cross-promotion strategies, potentially increasing sales across product lines.
5. **Dynamic Pricing:** Understanding the range of brands associated with a product can inform pricing strategies, potentially allowing for price adjustments based on perceived brand value.
6. **Product Development:** Insights into the attributes that drive recommendations

can inform future product development, ensuring new products are designed with recommendation potential in mind.

By using these marketing insights derived from our counterfactual analysis, businesses can improve their recommendation systems and enhance their overall marketing effectiveness.

1.2 Limitations and Challenges

1. **Computational Complexity:** The process of generating and evaluating counterfactual scenarios for each recommendation is computationally intensive, especially for large-scale knowledge graphs. This limits the real-time application of this method in production recommender systems.
2. **Attribute Independence Assumption:** Our analysis treats each attribute independently, which may not always reflect real-world scenarios where attributes can be correlated. This simplification might lead to some unrealistic counterfactual scenarios.
3. **Threshold Sensitivity:** The choice of threshold score for determining relevant counterfactuals can significantly impact the results. Finding the optimal threshold that balances inclusivity and relevance remains a challenge.
4. **Temporal Aspects:** Our analysis doesn't account for temporal changes in user preferences or product popularity, which could affect the relevance of counterfactual scenarios over time.
5. **Bias Reinforcement:** There's a risk that focusing on attributes that lead to recommendations could reinforce existing biases in the recommender system.